

Eightfold Assignment: Multi-Source Candidate Data Transformer

Design & Technical One-Pager | Candidate: Vivek (vivek@example.com)

1. Pipeline Architecture & Execution Flow

The engine operates on a decoupled seven-stage pipeline designed for modularity, resilience, and determinism:

- **Detect:** Identifies source types from input paths (CSV, ATS JSON, GitHub URL, Resume, notes).
- **Extract:** Instantiates respective readers (e.g. LLM-based Resume, Git API) wrapped in try/excepts to prevent failures.
- **Normalize:** Formats phone numbers to E.164, parses location strings to sub-objects, and resolves tech skills.
- **Merge:** Consolidates profiles using source priorities and completes arrays/sub-objects using union rules.
- **Confidence:** Calculates reliability per field based on completeness, agreements, and trusted source bonuses.
- **Project:** Transforms the unified canonical record into custom formats based on dynamic config paths.
- **Validate:** Enforces JSON schema constraints at both the canonical level (pre-projection) and configuration level.

2. Canonical Schema Definition

We enforce a strict Candidate Model structure to unify conflicting source types: **candidate_id** (SHA256 string), **full_name** (string), **emails** (list), **phones** (list, E.164 format), **location** (dict: city, region, country code), **links** (dict: linkedin, github, portfolio, other[]), **headline** (string), **years_experience** (number), **skills** (list of {name, confidence, sources}), **experience** (list of {company, title, start, end, summary}), **education** (list of {institution, degree, field, end_year}), and **provenance** / **overall_confidence** metadata.

3. Merge, Conflict Resolution & Confidence Model

Conflict Resolution: Single-value fields are resolved by sorting inputs by source priority first: *Resume (5)* > *LinkedIn (4)* > *GitHub (3)* > *ATS JSON (2)* > *Recruiter CSV/Notes (1)*, falling back to field completeness. List fields (emails, phones) are case-normalized and unioned. Complex objects (experience, education) are deduplicated by composite keys (company+title, institution+degree).

Confidence Scorer: Scores fields from 0.0 to 1.0 based on presence (+0.4), multi-source agreement (+0.3), critical fields full_name/emails (+0.2), and trusted sources resume/linkedin (+0.1). Overall confidence is the mean of field scores. Individual skills compute their own confidence: base rating (0.7-0.9 based on source trust) + 0.1 bonus per additional source.

4. Runtime Configuration Projection & Validation

A strict boundary separates the *canonical engine* from the *projection layer*. The canonical profile is validated against the default Candidate schema immediately after merge, before projection. `Projector.py` then traverses JSONPaths, renaming columns, applying E.164/canonical filters, and generating a dynamic JSON schema on the fly based on requested types, ensuring output validation is completely configurable without polluting the pipeline core.

5. Edge Cases & Descoped trade-offs

- **Unreliable LLM Parsing:** PDF/Word resume parsing uses Claude, falling back to custom regex parsers if API limits are hit.
- **India-centric Localizations:** Location parser splits and resolves Indian states, converting implicit regions to ISO country code 'IN'.
- **Descoped out-of-time:** Excluded complex graph-based skill taxonomy mapping, relying instead on cosine similarity matching against a hardcoded tech skill list.